

Applied Cryptography(CSE546)

End-Sem Examination

Total marks: 50

Name: _____

Roll No.: _____

Section A (5 marks)

1. A ciphertext was encrypted using the Vigenère cipher with an unknown key length L . The Index of Coincidence (IC) of the ciphertext is approximately 0.045. The IC of random text is ≈ 0.038 , and that of English text is ≈ 0.065 . What is the most reasonable conclusion?
- A. The key length is very short (e.g., 1 or 2).
 - B. The key length is very long (close to the ciphertext length).
 - C. The plaintext is just a repeated character.
 - D. The key length is likely between 5 and 10.

Solution: D

2. A one-time pad is used correctly: the key is truly random, as long as the message, never reused, and kept secret. Which statement is true even against an unbounded adversary?
- A. The ciphertext reveals half the bits of the plaintext on average.
 - B. The ciphertext can be decrypted to any possible plaintext of the same length with equal probability.
 - C. The ciphertext must be longer than the plaintext to achieve perfect secrecy.
 - D. The ciphertext alone can be used to recover the key with probability 1 given enough computation.

Solution: B

3. A 4-stage LFSR has a primitive feedback polynomial. The initial state is 1001. How many distinct states will the LFSR cycle through before repeating?
- A. 4
 - B. 8
 - C. 15
 - D. 16

Solution: C

4. Which of the following is a key structural difference between DES and AES?
- A. AES uses a Feistel network; DES uses a substitution-permutation network.
 - B. DES uses a Feistel network; AES uses a substitution-permutation network.
 - C. Both use Feistel networks, but AES has more rounds.
 - D. Both use substitution-permutation networks, but DES has a smaller block size.

Solution: B

5. A single-bit error occurs in a ciphertext block C_i during transmission. Which mode of operation will cause only the corresponding plaintext block P_i to be corrupted (and no other blocks)?
- A. CBC
 - B. CFB
 - C. s -bit CFB
 - D. OFB

Solution: D

Section B (45 marks)

6. Let $n = 33$ and $a = 10$.
- (a) Compute $\varphi(n)$.
 - (b) Verify Euler's theorem
 - (c) Find the smallest positive integer x such that $10^x \equiv 1 \pmod{33}$.

Solution:

- (a) We factor:

$$33 = 3 \times 11.$$

Hence,

$$\varphi(33) = \varphi(3 \times 11) = \varphi(3) \times \varphi(11) = 2 \times 10 = 20$$

- (b) We compute:

$$10^2 = 100 \equiv 100 - 3 \times 33 = 1 \pmod{33}.$$

Thus,

$$10^{20} = (10^2)^{10} \equiv 1^{10} = 1 \pmod{33}.$$

So Euler's theorem is verified.

- (c) We check powers:

$$10^1 = 10 \not\equiv 1 \pmod{33}, \quad 10^2 \equiv 1 \pmod{33}.$$

Hence the smallest such exponent is:

$$x = 2.$$

7. For the RSA Encryption Scheme let $p = 5$, $q = 7$, and the public exponent $e = 5$.
- (a) Compute n and $\varphi(n)$.
 - (b) Determine the private exponent d .

(c) Encrypt $m = 2$.

Solution:

(a)

$$n = 5 \cdot 7 = 35, \quad \varphi(n) = 4 \cdot 6 = 24.$$

(b) Solve $5d \equiv 1 \pmod{24}$:

$$5 \cdot 5 = 25 \equiv 1 \pmod{24} \Rightarrow d = 5.$$

(c)

$$c = 2^5 \bmod 35 = 32.$$

8. (a) State Fermat's Theorem

(b) Using Fermat's Theorem find the remainder when 2^{2023} is divided by 13.

Solution:

(a) If p is prime and n is a positive integer not divisible by p then

$$n^p \equiv n \pmod{p}$$

(b) Since 13 is prime and $\gcd(2, 13) = 1$, by Fermat's Little Theorem:

$$2^{13-1} \equiv 1 \pmod{13}.$$

Write:

$$2023 = 12 \cdot 168 + 7.$$

Thus,

$$2^{2023} = (2^{12})^{168} \cdot 2^7 \equiv 1^{168} \cdot 2^7 \pmod{13}.$$

Compute:

$$2^3 = 8 \pmod{13}, 2^4 = 16 \equiv 3 \pmod{13}.$$

$$2^7 = 2^4 \cdot 2^3 \equiv 3 \cdot 8 = 24 \equiv 11 \pmod{13}.$$

Hence,

$$2^{2023} \equiv 11 \pmod{13}.$$

9. Apply the Extended Euclidean Algorithm to find the greatest common divisor of 83 and 29, and express it as a linear combination: $83u + 29v = \gcd(83, 29)$. Hence find the inverse of 29 modulo 83.

Solution:

Euclidean algorithm:

$$\begin{aligned} 83 &= 2 \cdot 29 + 25 & (2 \cdot 29 = 58, 83 - 58 = 25) \\ 29 &= 1 \cdot 25 + 4 \\ 25 &= 6 \cdot 4 + 1 \\ 4 &= 4 \cdot 1 + 0 \end{aligned}$$

Thus $\gcd(83, 29) = 1$.

Back-substitution:

$$\begin{aligned} 1 &= 25 - 6 \cdot 4 \\ 4 &= 29 - 1 \cdot 25 \Rightarrow 1 = 25 - 6(29 - 25) = 7 \cdot 25 - 6 \cdot 29 \\ 25 &= 83 - 2 \cdot 29 \Rightarrow 1 = 7(83 - 2 \cdot 29) - 6 \cdot 29 = 7 \cdot 83 - 14 \cdot 29 - 6 \cdot 29 = 7 \cdot 83 - 20 \cdot 29 \end{aligned}$$

Hence $83(7) + 29(-20) = 1$.

The inverse of 29 modulo 83 is the coefficient of 29 when the equation is written as $29 \cdot v \equiv 1$. From $7 \cdot 83 - 20 \cdot 29 = 1$ we have $-20 \cdot 29 \equiv 1 \pmod{83}$, so the inverse is $-20 \bmod 83 = 63$.

10. (a) What is a cyclic group? What is the *Order* of a cyclic group?
 (b) Define Discrete Logarithm Problem
 (c) Show that ElGamal is *multiplicatively homomorphic*

Solution:

- (a) A group G is cyclic if there exists an element $g \in G$ called a generator, such that every element of G can be written as a power of g .

$$G = \langle g \rangle = \{g^0, g^1, g^2, \dots, g^{q-1}\}$$

The *Order* of G is the smallest positive integer k such that $g^k = e$.

- (b) Let $G = \langle g \rangle$ be a cyclic group of order q . Given $h \in G$, the discrete logarithm of h with respect to g is the unique integer $x \in \{0, 1, \dots, q-1\}$ such that $g^x = h$. The Discrete Logarithm Problem, states that given g and g^x , compute x .
 (c) ElGamal is multiplicatively homomorphic: given two ciphertexts, one can compute an encryption of the product of the plaintexts without decrypting.

Let

$$c_1 = \text{Enc}(pk, m_1) = (g^{r_1}, m_1 h^{r_1}) \quad \text{and} \quad c_2 = \text{Enc}(pk, m_2) = (g^{r_2}, m_2 h^{r_2}).$$

Then:

$$c_1 \otimes c_2 := (g^{r_1} \cdot g^{r_2}, m_1 h^{r_1} \cdot m_2 h^{r_2}) = (g^{r_1+r_2}, (m_1 m_2) h^{r_1+r_2}) = \text{Enc}(pk, m_1 \cdot m_2).$$

11. Alice and Bob agree to use the prime $p = 23$ and the primitive root $g = 5$. Alice chooses a secret integer $a = 6$ and Bob chooses a secret integer $b = 15$.
 (a) Compute Alice's public value and Bob's public value.
 (b) Compute the shared secret key that Alice would compute, and verify that Bob obtains the same value.

Solution:

Step 1: Compute Alice's public value A

$$A = 5^6 \bmod 23$$

Compute step by step modulo 23:

$$5^2 = 25 \equiv 2 \pmod{23}$$

$$5^4 = (5^2)^2 \equiv 2^2 = 4 \pmod{23}$$

$$5^6 = 5^4 \cdot 5^2 \equiv 4 \cdot 2 = 8 \pmod{23}$$

Thus $A = 8$.

Step 2: Compute Bob's public value B

$$B = 5^{15} \bmod 23$$

$$5^6 \equiv 8 \pmod{23} \quad (\text{from above})$$

$$5^{12} = (5^6)^2 \equiv 8^2 = 64 \equiv 64 - 2 \cdot 23 = 64 - 46 = 18 \pmod{23}$$

$$5^{15} = 5^{12} \cdot 5^3 \equiv 18 \cdot 5^3$$

Now $5^3 = 125$. $125 - 5 \cdot 23 = 125 - 115 = 10$, so $5^3 \equiv 10 \pmod{23}$. Thus $5^{15} \equiv 18 \cdot 10 = 180$. Reduce modulo 23: $23 \cdot 7 = 161$, $180 - 161 = 19$. So $B = 19$.

Alternatively, one could compute directly using repeated squaring:

$$5^{15} = 5^8 \cdot 5^4 \cdot 5^2 \cdot 5^1$$

But the above is fine.

Step 3: Alice computes the shared secret

Alice receives Bob's public value $B = 19$. She computes:

$$K = B^a \bmod p = 19^6 \bmod 23$$

Compute $19 \equiv -4 \pmod{23}$. Then:

$$(-4)^2 = 16 \pmod{23}$$

$$(-4)^4 = 16^2 = 256 \equiv 256 - 11 \cdot 23 = 256 - 253 = 3 \pmod{23}$$

$$(-4)^6 = (-4)^4 \cdot (-4)^2 \equiv 3 \cdot 16 = 48 \equiv 48 - 2 \cdot 23 = 2 \pmod{23}$$

Thus Alice's shared secret is $K = 2$.

Step 4: Bob computes the shared secret

Bob receives Alice's public value $A = 8$. He computes:

$$K = A^b \bmod p = 8^{15} \bmod 23$$

We already have $8^2 = 64 \equiv 18 \pmod{23}$, $8^4 \equiv 18^2 = 324 \equiv 324 - 14 \cdot 23 = 324 - 322 = 2 \pmod{23}$, $8^8 \equiv 2^2 = 4 \pmod{23}$. Now $15 = 8 + 4 + 2 + 1$:

$$8^{15} = 8^8 \cdot 8^4 \cdot 8^2 \cdot 8^1 \equiv 4 \cdot 2 \cdot 18 \cdot 8 \pmod{23}$$

Compute stepwise: $4 \cdot 2 = 8$, $8 \cdot 18 = 144$, $144 \bmod 23$: $23 \cdot 6 = 138$, remainder 6. Then $6 \cdot 8 = 48 \equiv 2 \pmod{23}$. So Bob also obtains $K = 2$.

The shared secret key is 2.

12. (a) Write the formal description of the ElGamal Signature Scheme, including Setup, Signing, and Verification algorithms.
- (b) In case of the textbook RSA signature scheme, consider you are provided with the signature of a message $m_1 = 3$, as $\text{Sign}(3) = 3^2 \bmod 5$. Show that this information can allow an adversary to construct a valid signature of some other message.

Solution:**(a) Setup**

- Public parameters: large prime p , generator g of Z_p^* .
- Private key: random $x \in \{1, \dots, p-2\}$.
- Public key: $y = g^x \bmod p$.

Signing

1. Choose a **fresh, secret, random** k with $\gcd(k, p-1) = 1$.
2. Compute $r = g^k \bmod p$.
3. Compute $s = k^{-1}(H(m) - xr) \bmod (p-1)$.
4. Signature: $\sigma = (r, s)$.

Verification

Check whether:

$$g^{H(m)} \equiv y^r \cdot r^s \pmod{p}.$$

Accept if equality holds; reject otherwise.

- (b) This was a trick question. The question had a mistake in it. It is necessary for RSA scheme to have $n = p \times q$ where p and q are primes. However in the question $n = 5$ does not satisfy this property.

The correct answer should have been - This is not RSA signature scheme.

13. (a) What are the three classical security properties that should be satisfied by a cryptographic hash function? Describe each formally.
- (b) What are the four distinct auxiliary round functions used in SHA-1? Also write the round constants.

Solution:

- (a) There are three classical security properties:

Preimage Resistance (One-wayness): Given a digest $y = H(x)$, it is computationally infeasible to find *any* x' such that $H(x') = y$.

Second Preimage Resistance (Weak Collision Resistance): Given a message x , it is computationally infeasible to find $x' \neq x$ such that $H(x') = H(x)$.

Collision Resistance (Strong Collision Resistance): It is computationally infeasible to find *any* pair (x, x') with $x \neq x'$ and $H(x) = H(x')$.

- (b) The functions are: $(B \& C) \mid (\neg B \& D)$, $B \oplus C \oplus D$, $(B \& C) \mid (B \& D) \mid (C \& D)$ and $B \oplus C \oplus D$.

The constants are $\lfloor 2^{30}\sqrt{2} \rfloor$, $\lfloor 2^{30}\sqrt{3} \rfloor$, $\lfloor 2^{30}\sqrt{5} \rfloor$, $\lfloor 2^{30}\sqrt{10} \rfloor$ respectively

14. (a) Give a formal definition of a Public Key Cryptosystem (PKC) as a triple of algorithms, and state the correctness condition.
 (b) Describe how PKC can be used to achieve confidentiality and authenticity.
 (c) List two real-world applications of public key cryptography.

Solution:

- (a) A Public Key Cryptosystem (PKC) is defined as a triple of probabilistic polynomial-time algorithms

(KeyGen, Enc, Dec)

such that:

- **KeyGen**(1^λ): takes a security parameter λ and outputs a pair (pk, sk) , where pk is the public key and sk is the secret key.
- **Enc**(pk, m): takes the public key pk and a message m from the message space, and outputs a ciphertext c .
- **Dec**(sk, c): takes the secret key sk and a ciphertext c , and outputs a message m or a special symbol $\perp$ indicating failure.

Correctness condition: For all security parameters λ , all key pairs (pk, sk) generated by **KeyGen**(1^λ), and all messages m in the message space, it holds that:

$$\text{Dec}(sk, \text{Enc}(pk, m)) = m$$

with probability 1 (or with overwhelming probability if the algorithms are randomized).

- (b) To achieve confidentiality and authenticity simultaneously, we combine encryption and digital signatures:
- The sender first signs the message m using their secret key sk_S , producing a signature σ .
 - Then the sender encrypts the pair (m, σ) using the recipient's public key pk_R , obtaining ciphertext c .
 - The sender sends c to the recipient.

At the receiver's side:

- The recipient decrypts c using their secret key sk_R to recover (m, σ) .
- The recipient verifies σ using the sender's public key pk_S .

This ensures:

- **Confidentiality:** Only the intended recipient can decrypt the message.
- **Authenticity & Integrity:** The valid signature guarantees the message came from the sender and was not modified.

Note: Answer can also be shown using a diagram.

(c) Two real-world applications of public key cryptography are (Any other related example will also work.):

- HTTPS/TLS for secure web communication.
- PGP/GPG for secure email encryption and signing.